

What Is Javascript

Javascript (Js) Is A **Light-Weight Object-Oriented Programming Language** Which Is Used By **Several Websites For Scripting The Webpages**. It Is An Interpreted, Full-Fledged Programming Language **That Enables Dynamic Interactivity On Websites When Applied To An HTML Document**. It Was Introduced In The Year 1995 For Adding Programs To The Webpages In The Netscape Navigator Browser. Since Then, It Has Been Adopted By All Other Graphical Web Browsers. With Javascript, Users Can Build Modern Web Applications To Interact Directly Without Reloading The Page Every Time. The Traditional Website Uses Js To Provide Several Forms Of Interactivity And Simplicity.

Although, Javascript Has No Connectivity With Java Programming Language. The Name Was Suggested And Provided In The Times When Java Was Gaining Popularity In The Market. In Addition To Web Browsers, Databases Such As Couchdb And MongoDB Uses Javascript As Their Scripting And Query Language.

Features Of Javascript

There Are Following Features Of Javascript:

1. All Popular Web Browsers Support Javascript As They Provide Built-In Execution Environments.
2. Javascript Follows The Syntax And Structure Of The C Programming Language. Thus, It Is A Structured Programming Language.
3. Javascript Is A Weakly Typed Language, Where Certain Types Are Implicitly Cast (Depending On The Operation).
4. Javascript Is An Object-Oriented Programming Language That Uses Prototypes Rather Than Using Classes For Inheritance.
5. It Is A Light-Weighted And Interpreted Language.
6. It Is A Case-Sensitive Language.
7. Javascript Is Supportable In Several Operating Systems Including, Windows, MacOS, Etc.
8. It Provides Good Control To The Users Over The Web Browsers.

Common Event Listeners

Following Is The List Of Javascript Listeners' Definitions Which Have Specific Types Of Methods.

- **Onload:** When Your Page Loads, It Performs Accordingly.
- **Onclick:** When A User Clicks On A Button Or Inputs It Occurs.
- **Onmouseover:** When A User Mouses Over On The Button.
- **Onfocus:** Certain Scenarios When A User Keeps The Cursor In A Form Field.
- **Onblur:** If A Particular Form Field Leaves Within It.

Benefits Of Learning Javascript

Web Development

Desktop App Development

Mobile App Development

- Web Development Use
 - JQuery
 - Angular Js
 - React Js
 - Vue Js
 - Node Js
- Desktop App Development
 - Electron Js
- Mobile App Development
 - React Native
 - Angular Js
 - React Js
 - Vue Js
 - Node Js

Use Of Javascript In Web Development

- Dropdown Menu
- Video players
- Animated slider
- Zoom effect
- Map
- Animated gallery
- Chat graphs
- Form validation
- Pop up window

- Audio Players
- Calendar

2 way of java script

Inpage java script

External java script

Inpage java script

```
<html >
<head>

  <title>JavaScript</title>
  <!-- <script>
    document.write("Hello from Yahoo Baba");
  </script> -->
</head>
<body>
  <h1>Hello</h1>
  <script>
    document.write("Hello from Yahoo Baba");
  </script>

</body>
</html>
```

External java script

external JavaScript file must be saved by .js extension. It is recommended to embed all JavaScript files into a single file. It increases the speed of the webpage.

□

□

□ **Example**

<script>

document.write("JavaScript is a simple language ");

</script>

Between the body tag of html

Between the head tag of html

In .js file (external javaScript)

External scripts are practical when the same code is used in many different web pages.

JavaScript files have the file extension **.js**.

To use an external script, put the name of the script file in the src (source) attribute of a <script> tag:

Example

```
<html lang="en">
<head>

  <title>JavaScript</title>
  <!-- External JavaScript -->
  <script src="js/test.js"></script>
</head>
<body>
  <h1>Hello</h1>
</body>
</html>
```

HTML Tags In Javascript

In this example, the JavaScript code uses the **alert** function to display a pop-up message saying "Hello from JavaScript!".

```
<script>
  document.write("Hello World <br><br>");
  document.write("<i><b>Hello java script<b></i>");
</script>
```

JS Comments

Advantages of JavaScript comments

To make code easy to understand

To avoid the unnecessary code

Types of JavaScript Comments

There are two types of comments in JavaScript.

1. Single-line Comment
2. Multi-line Comment

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta http-equiv="X-UA-Compatible" content="ie=edge">
  <title>JavaScript</title>
  <script>
```

```
document.write("Hello World <br><br>"); // JavaScript Single line Comment
```

```
document.write("<i><b>Hello from Yahoo Baba<b></i>"); /* JavaScript Multi line Comment
```

It can be used to add single as well as multi line comments. So, it is more convenient.*/

```
</script>
</head>
<body>
  <h1>Yahoo Baba</h1>
</body>
</html>
```

JS Variables

Variables are Containers for Storing Data

JavaScript Variables can be declared in 4 ways:

- Automatically
- Using **var**
- Using **let**
- Using **const**

Using **var**

Ex.

```
<script>
  var x = "Hello";
  var y = "World";
  // x = WOW
  X= 111
  document.write(x + y);
</script>
```

Write a way of variable

Firstname

First_name

firstname

firstname99

wrong way of variable

first name

99firstname

- Using **let**

Variables defined with **let** cannot be **Redeclared**

Variables defined with **let** must be **Declared** before use

- Using **const**

Variables defined with **const** cannot be **Redeclared**

Variables defined with **const** cannot be **Reassigned**

Var	Let	Const
<pre>var x = "Hello"; var x = "World"; x = "WoW";</pre>	<pre>let x = "Hello"; let x = "World"; x = "WoW";</pre>	<pre>const x = "Hello"; const x = "World"; x = "WoW";</pre>

JS Data Types

JavaScript provides different **data types** to hold different types of values. There are two types of data types in JavaScript.

1. Primitive data type
2. Non-primitive (reference) data type

JavaScript is a **dynamic type language**, means you don't need to specify type of the variable because it is dynamically used by JavaScript engine. You need to use **var** here to specify the data type. It can hold any type of values such as numbers, strings etc. For example:

1. `var a=40;`//holding number
2. `var b="Rahul";`//holding string

<code>var x = "Hello World";</code>	→	String
<code>var x = 25;</code>	→	Number
<code>var x = true;</code>	→	Boolean
<code>var x = ["HTML", "CSS", "JS"];</code>	→	Array
<code>var x = {first:"Jane", last:"Doe"};</code>	→	Object
<code>var x = null;</code>	→	Null
<code>var x;</code>	→	Undefined

```
/* String Data Type */  
var a = "Hello World";  
document.write(a);  
document.write("<br>");  
  
document.write(typeof a);
```

```
/* Number Data Type */  
var b = 25;  
document.write(b);  
document.write("<br>");  
document.write(typeof b);  
  
document.write("<br><br>");
```

```
/* Boolean Data Type */  
var c = true;  
document.write(c);  
document.write("<br>");  
document.write(typeof c);  
document.write("<br><br>");
```

```
/* Array Data Type */  
var d= ["HTML", "CSS", "JS"];  
document.write(d);  
document.write("<br>");  
document.write(typeof d);  
document.write("<br><br>");
```

```
/* Object Data Type */  
var x= {first:"Jane",last:"Doe"};  
document.write(x);  
document.write("<br>");  
document.write(typeof x);  
document.write("<br><br>");
```

```
/* Null Data Type */  
var y = null;  
document.write(y);
```

```
document.write("<br>");
document.write(typeof y);
document.write("<br><br>");
```

```
/* undefined Data Type */
var z;
document.write(z);
document.write("<br>");
document.write(typeof z);
```

```
document.write("<br><br>");
```

Operators

1. JavaScript Arithmetic Operators

Operator	Description	Example
+	Addition	10+20 = 30
-	Subtraction	20-10 = 10
*	Multiplication	10*20 = 200
/	Division	20/10 = 2
%	Modulus (Remainder)	20%10 = 0
++	Increment	var a=10; a++; Now a = 11
--	Decrement	var a=10; a--; Now a = 9

EX.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<meta http-equiv="X-UA-Compatible"
content="ie=edge">
<title>JavaScript</title>
<script>
/* Addition Arithmetic Operators */
var a = 10;
var b = 10;
var c = a + b;
document.write(c);
document.write("<br><br>");
```



```
/* Subtraction Arithmetic Operators */
var m = 10;
var n = 5;
var o = m - n;
document.write(o);
document.write("<br><br>");

/* Multiplication Arithmetic Operators */
var p = 10;
var q = 10;
var r = a * b;
document.write(r);
document.write("<br><br>");

/* Exponentiation Arithmetic Operators */
var d = 10;
var e = 3;
var f = d ** e; //10*10*10=1000
document.write(f);
document.write("<br><br>");

/* Division Arithmetic Operators */
var s = 14;
var t = 2;
var w = s / t;
document.write(w);
document.write("<br><br>");

/* Modulus(Remainder) Arithmetic Operators */
var x = 14;
var y = 2;
var z = a % b; // 2/14 7*2=14 14-14=0
document.write(z);
document.write("<br><br>");

/* Increment Arithmetic Operators */
var i = 10;
var j = 3;
document.write(i + j);
document.write("<br>");
i++; // a=a+1
document.write(i + j);
document.write("<br><br>");
```

```

/* Decrement Arithmetic Operators */
var g = 10;
var h = 3;
document.write(g + h);
document.write("<br>");
g--;
document.write(g + h);

```

```

/* Decrement Arithmetic Operators */
var a = 10;
var b = 3;
var c = a+b*2
//var c= (a+b)*2
document.write(c);

```

2. JavaScript assignment operator

JavaScript **assignment operator** is **equal (=)** which assigns the value of the right-hand operand to its left-hand operand. That is if $a = b$ assigns the value of b to a .

The simple assignment operator is used to assign a value to a variable

Operator	Example	Same As
=	$x = y$	$x = y$
+=	$x += y$	$x = x + y$
-=	$x -= y$	$x = x - y$
*=	$x *= y$	$x = x * y$
/=	$x /= y$	$x = x / y$
%=	$x \% = y$	$x = x \% y$
**=	$x ** = y$	$x = x ** y$

Ex.

```

/* Equalto Assignment Operators */

var m = 10;
var n = 3;
o = m + n;
document.write(o);
document.write('<br><br>');

```

```
/* Addition Assignment Operators */  
    var a = 10;  
    var b = 3;  
    a += b;  
    document.write(a);  
    document.write('<br><br>');
```

JS With Google Chrome Console

```
<script>  
    var x = 50;  
    console.log(x);  
  
    console.log([1,2,3]);  
  
    console.table([1, 2, 3]);  
  
    console.error("Something went wrong.");  
  
    console.time("Test");  
    console.warn("This is just warning");  
    console.timeEnd("Test");  
    //console.clear();  
</script>
```

Comparison operators s

JavaScript **Comparison operators** are mainly used to perform the logical operations that determine the equality or difference between the values. **Comparison operators** are used in logical expressions to determine their equality or differences in variables or values.

JavaScript **Comparison Operators list:** There are so many comparison operators as shown in the table with the description.

OPERATOR NAME	USAGE	OPERATION
Equality Operator	<code>a==b</code>	Compares the equality of two operators
Inequality Operator	<code>a!=b</code>	Compares inequality of two operators
Strict Equality Operator	<code>a===b</code>	Compares both value and type of the operand
Strict Inequality Operator	<code>a!==b</code>	Compares inequality with type
Greater than Operator	<code>a>b</code>	Checks if the left operator is greater than the right operator
Greater than or equal Operator	<code>a>=b</code>	Checks if the left operator is greater than or equal to the right operator
Less than Operator	<code>a<b</code>	Checks if the left operator is smaller than the right operator
Less than or equal Operator	<code>a<=b</code>	Checks if the left operator is smaller than or equal to the right operator

Ex.

`<script>`

```
/* Equal to Comparison Operators */
```

```
var a = 10;
```

```
var b = 20;
```

```
document.write( a == b);
```

```
document.write('<br><br>');
```

```
/* Equal value and equal type Comparison Operators */
```

```
var c = 10;
```

```
var d= "10";
```

```
document.write(a === b);
```

```
document.write('<br><br>');
```

```
/* Not Equal Comparison Operators */
```

```
var e = 2;
```

```
var f = 3;
```

```
document.write(e != f);
```

```
document.write('<br><br>');
```

```
/* Not Equal value or not equal type Comparison Operators */
```

```
var g = 2;
```

```
var h = '2';
```

```
document.write(g !== h);
```

```
document.write('<br><br>');
```

```
/* Greater Than Comparison Operators */
var i = 2;
var j = 4;
document.write(i > j);
    document.write('<br><br>');

/* Less Than Comparison Operators */
var m = 10;
var n = 20;
document.write(m < n);
    document.write('<br><br>');

/* Greater Than or Equal To Comparison Operators */
var p = 10;
var q = 20;
document.write(p >= q);
    document.write('<br><br>');

/* Less Than or Equal To Comparison Operators */
var x= 20;
var y = 20;
document.write(x <= y);
    document.write('<br><br>');
</script>
```

JavaScript Output

JavaScript Display Possibilities

JavaScript can "display" data in different ways:

Writing into an HTML element, using **innerHTML**.

Writing into the HTML output using **document.write()**.

Writing into an alert box, using **window.alert()**.

Writing into the browser console, using **console.log()**.

Ex. **document.write()**.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h1>My First Web Page</h1>
```

```
<p>My first paragraph.</p>
```

```
<script>
document.write(5 + 6);
</script>
```

```
</body>
</html>
```

window.alert().

```
<button type="button" onclick="document.write(5 + 6)">Try it</button>
```

innerHTML.

```
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML = 5 + 6;
</script>
```

using console.log().

```
<script>
console.log(5 + 6);
</script>
```

JavaScript Function

JavaScript functions are used to perform operations. We can call JavaScript function many times to reuse the code.

Advantage of JavaScript function

There are mainly two advantages of JavaScript functions.

1. **Code reusability:** We can call a function several times so it save coding.
2. **Less coding:** It makes our program compact. We don't need to write many lines of code each time to perform a common task.

JavaScript Function Syntax

```
function functionName([arg1, arg2, ...argN])
{
    //code to be executed
}
```

EX.

```
<script>
  function hello() {
    document.write("Hello Everybody");
  }
  Function world() {
    document.write(" Me BHARAT CHAUDHARY");
  }

  hello();
  document.write("<br>");
  world();
  document.write("<br>");
  hello();
  document.write("<br>");
  hello();
</script>
```